# Chapter 10: Exponential Functions (Hàm số Mũ)

## 📖 Chapter Overview

Exponential functions are among the most important mathematical models in science, economics, and everyday life. They describe phenomena involving rapid growth or decay, from population dynamics and compound interest to radioactive decay and viral spread. In this chapter, we explore the definition, properties, and graphs of exponential functions, building on our understanding of exponents with real numbers. We'll discover why these functions are so powerful and how they connect to real-world applications.

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## 📚 Bilingual Vocabulary (Từ vựng Song ngữ)

### Basic Concepts (Khái niệm Cơ bản)

| English | IPA | Vietnamese | Example |
|---|---|---|---|
| **Exponential function** | /ˌekspəˈnenʃəl ˈfʌŋkʃən/ | Hàm số mũ | The function $y = 2^x$ is an exponential function. |
| **Base** | /beɪs/ | Cơ số | In $y = 3^x$, the base is 3. |
| **Exponent** | /ɪkˈspoʊnənt/ | Số mũ | In $2^5$, the exponent is 5. |
| **Power** | /ˈpaʊər/ | Lũy thừa | $2^3 = 8$ is a power of 2. |
| **Exponential growth** | /ˌekspəˈnenʃəl groʊθ/ | Tăng trưởng mũ | Population often exhibits exponential growth. |

| | | | |
|---|---|---|---|
| **Exponential decay** | /ˌekspəˈnenʃəl dɪˈkeɪ/ | Suy giảm mũ | Radioactive substances undergo exponential decay. |

## Properties and Behavior (Tính chất và Hành vi)

| English | IPA | Vietnamese | Example |
|---|---|---|---|
| **Domain** | /doʊˈmeɪn/ | Tập xác định | The domain of $y = a^x$ is all real numbers. |
| **Range** | /reɪndʒ/ | Tập giá trị | The range of $y = a^x$ is $(0, +\infty)$. |
| **Increasing function** | /ɪnˈkriːsɪŋ ˈfʌŋkʃən/ | Hàm đồng biến | When $a > 1$, $y = a^x$ is an increasing function. |
| **Decreasing function** | /dɪˈkriːsɪŋ ˈfʌŋkʃən/ | Hàm nghịch biến | When $0 < a < 1$, $y = a^x$ is a decreasing function. |
| **Asymptote** | /ˈæsɪmptoʊt/ | Tiệm cận | The x-axis is a horizontal asymptote of $y = a^x$. |

| | | | |
|---|---|---|---|
| **Continuous** | /kənˈtɪnjuəs/ | Liên tục | Exponential functions are continuous everywhere. |

### Special Numbers and Applications (Số Đặc biệt và Ứng dụng)

| English | IPA | Vietnamese | Example |
|---|---|---|---|
| **Natural base e** | /ˈnætʃərəl beɪs iː/ | Cơ số tự nhiên e | $e \approx 2.71828$ is the natural base. |
| **Compound interest** | /ˈkɑːmpaʊnd ˈɪntrəst/ | Lãi kép | Banks calculate compound interest using exponential functions. |
| **Growth rate** | /groʊθ reɪt/ | Tỉ lệ tăng trưởng | A 5% growth rate means $r = 0.05$ . |
| **Initial value** | /ɪˈnɪʃəl ˈvæljuː/ | Giá trị ban đầu | $P$ represents the initial value in $A = Pe^{rt}$ . |
| **Half-life** | /ˈhæf laɪf/ | Chu kỳ bán rã | Carbon-14 has a half-life of about 5,730 years. |

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## 10.1 Definition and Recognition (Định nghĩa và Nhận dạng)

### Definition of Exponential Function

> **Definition:** Let
> $a$

> be a positive real number with
> $a \neq 1$
> . The function
> $f(x) = a^x$
> is called an **exponential function with base**
> $a$
> .

**Key points:**

- The base
  $a$
  must be positive:
  $a > 0$
- The base cannot equal 1:
  $a \neq 1$
  (because
  $1^x = 1$
  for all
  $x$
  , which is a constant function)
- The exponent
  $x$
  can be any real number
- The output
  $a^x$
  is always positive

## Recognizing Exponential Functions

An exponential function has the variable in the exponent, not in the base.

**Exponential functions:**

- $y = 2^x$
  (base 2)
- $y = (\frac{1}{3})^x$
  (base

$\frac{1}{3}$
)

- $y = e^x$
  (base
  $e$
  )
- $y = 5 \cdot 3^x$
  (exponential with coefficient)

**NOT exponential functions:**

- $y = x^2$
  (power function, variable is the base)
- $y = x^{-1}$
  (power function)
- $y = 1^x = 1$
  (constant function, base equals 1)

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## 10.2 Properties of Exponential Functions (Tính chất của Hàm số Mũ)

### General Properties

For the exponential function
$y = a^x$
where
$a > 0$
and
$a eq 1$
:

> **Properties:**
>
> 1. **Domain:**
>    R
>    (all real numbers)
> 2. **Range:**
>    $(0, +\infty)$

> (all positive real numbers)
>
> 3. **Continuity:** The function is continuous on $\mathbb{R}$
>
> 4. **Intercepts:**
>    - y-intercept: $(0, 1)$ because $a^0 = 1$
>    - No x-intercept (the graph never crosses the x-axis)
>
> 5. **Horizontal asymptote:** The x-axis ( $y = 0$ ) is a horizontal asymptote

## Monotonicity (Behavior)

The behavior of $y = a^x$ depends on the value of the base $a$ :

**Case 1: When $a > 1$**

- The function is **increasing** (strictly monotone increasing) on $\mathbb{R}$
- As $x \to +\infty$ , we have $a^x \to +\infty$ (grows without bound)
- As $x \to -\infty$ , we have

$a^x \to 0^+$ (approaches zero from above)

- Larger values of $a$ produce steeper growth

**Case 2: When** $0 < a < 1$

- The function is **decreasing** (strictly monotone decreasing) on $\mathbb{R}$
- As $x \to +\infty$, we have $a^x \to 0^+$ (approaches zero from above)
- As $x \to -\infty$, we have $a^x \to +\infty$ (grows without bound)
- This represents exponential decay

## Important Inequality

For exponential functions with base $a > 1$:

$$a^{x_1} < a^{x_2} \quad \text{if and only if} \quad x_1 < x_2$$

For exponential functions with base $0 < a < 1$:

$$a^{x_1} > a^{x_2} \quad \text{if and only if} \quad x_1 < x_2$$

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## 10.3 Graphs of Exponential Functions (Đồ thị Hàm số Mũ)

### Graph of $y = a^x$ when $a > 1$

The graph has these characteristics:

- Passes through the point $(0, 1)$
- Passes through the point $(1, a)$
- Increases from left to right
- Lies entirely above the x-axis
- Approaches the x-axis as $x \to -\infty$
- Rises steeply as $x \to +\infty$

**Table of values for** $y = 2^x$:

| $x$ | $-3$ | $-2$ | $-1$ | $0$ | $1$ | $2$ | $3$ |
|---|---|---|---|---|---|---|---|
| $y$ | $\frac{1}{8}$ | $\frac{1}{4}$ | $\frac{1}{2}$ | $1$ | $2$ | $4$ | $8$ |

### Graph of $y = a^x$ when

## $0 < a < 1$

The graph has these characteristics:

- Passes through the point $(0, 1)$
- Passes through the point $(1, a)$
- Decreases from left to right
- Lies entirely above the x-axis
- Approaches the x-axis as $x \to +\infty$
- Rises steeply as $x \to -\infty$

**Table of values for** $y = (\frac{1}{2})^x$:

| $x$ | $-3$ | $-2$ | $-1$ | $0$ | $1$ | $2$ | $3$ |
|---|---|---|---|---|---|---|---|
| $y$ | $8$ | $4$ | $2$ | $1$ | $\frac{1}{2}$ | $\frac{1}{4}$ | $\frac{1}{8}$ |

### Symmetry Property

The graphs of $y = a^x$ and $y = (\frac{1}{a})^x$ are symmetric with respect to the y-axis because: $(\frac{1}{a})^x = a^{-x}$

This means the graph of
$y = a^{-x}$
is the reflection of
$y = a^x$
across the y-axis.

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## 10.4 The Natural Exponential Function (Hàm số Mũ Tự nhiên)

### The Number e

The number
$e$
is one of the most important constants in mathematics, defined as:
$e = \lim_{n\to\infty} (1 + \frac{1}{n})^n \approx 2.71828$

This limit arises naturally in many contexts, particularly in continuous compound interest and calculus.

### The Function $y = e^x$

The natural exponential function
$y = e^x$
has all the properties of exponential functions with base
$a > 1$
:

- Domain:
  $\mathbb{R}$
- Range:
  $(0, +\infty)$
- Strictly increasing on
  $\mathbb{R}$
- Continuous everywhere

- Passes through $(0, 1)$

**Why is $e$ special?**
The function $y = e^x$ has a unique property: its rate of change at any point equals its value at that point. This makes it fundamental in calculus and differential equations.

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## 📝 Worked Examples (Ví dụ Mẫu)

### Example 1: Identifying Exponential Functions

**Problem:** Which of the following are exponential functions? For those that are, identify the base.

a) $y = 3^x$
b) $y = x^3$
c) $y = (\sqrt{2})^x$
d) $y = (-2)^x$
e) $y = \pi^x$

**Solution:**

a) $y = 3^x$ is an exponential function with base $a = 3$.

b) $y = x^3$

is NOT an exponential function. This is a power function where the variable is the base, not the exponent.

c)
$y = (\sqrt{2})^x$
is an exponential function with base
$a = \sqrt{2} \approx 1.414$
.

d)
$y = (-2)^x$
is NOT an exponential function because the base is negative. Exponential functions require
$a > 0$
.

e)
$y = \pi^x$
is an exponential function with base
$a = \pi \approx 3.14159$
.

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## Example 2: Evaluating Exponential Expressions

**Problem:** Calculate the following without a calculator:

a)
$8^{2/3}$
b)
$16^{-0.75}$
c)
$27^{1/3} + 81^{-0.75}$

**Solution:**

a)
$8^{2/3} = (8^{1/3})^2 = (\sqrt[3]{8})^2 = 2^2 = 4$

Alternatively:
$8^{2/3} = (2^3)^{2/3} = 2^{3 \cdot \frac{2}{3}} = 2^2 = 4$

b)

$16^{-0.75} = 16^{-3/4} = \frac{1}{16^{3/4}} = \frac{1}{(16^{1/4})^3} = \frac{1}{(\sqrt[4]{16})^3} = \frac{1}{2^3} = \frac{1}{8}$

c)

$27^{1/3} + 81^{-0.75} = \sqrt[3]{27} + 81^{-3/4}$
$= 3 + \frac{1}{81^{3/4}} = 3 + \frac{1}{(81^{1/4})^3} = 3 + \frac{1}{(\sqrt[4]{81})^3} = 3 + \frac{1}{3^3} = 3 + \frac{1}{27} = \frac{82}{27}$

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### Example 3: Comparing Exponential Values

**Problem:** Without using a calculator, compare the following pairs of numbers:

a)
$2^{3.5}$
and
$2^{3.7}$

b)
$(\frac{1}{3})^{0.5}$
and
$(\frac{1}{3})^{0.7}$

c)
$3^{\sqrt{2}}$
and
$5^{\sqrt{2}}$

**Solution:**

a) Since
$2 > 1$
and
$3.5 < 3.7$
, we have
$2^{3.5} < 2^{3.7}$
.

For exponential functions with base greater than 1, the function is increasing.

b) Since
$0 < \frac{1}{3} < 1$
and

$0.5 < 0.7$
, we have
$\left(\frac{1}{3}\right)^{0.5} > \left(\frac{1}{3}\right)^{0.7}$
.

For exponential functions with base between 0 and 1, the function is decreasing.

c) Both expressions have the same exponent
$\sqrt{2}$
. Since
$3 < 5$
and
$\sqrt{2} > 0$
, we have
$3^{\sqrt{2}} < 5^{\sqrt{2}}$
.

When comparing
$a^x$
and
$b^x$
with the same positive exponent, the larger base produces the larger result.

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### Example 4: Sketching Exponential Graphs

**Problem:** Sketch the graph of
$y = 3^x$
and identify key features.

**Solution:**

**Step 1:** Create a table of values.

| $x$ | $-2$ | $-1$ | $0$ | $1$ | $2$ |
|---|---|---|---|---|---|
| $y$ | $\frac{1}{9} \approx 0.11$ | $\frac{1}{3} \approx 0.33$ | $1$ | $3$ | $9$ |

**Step 2:** Plot the points and draw a smooth curve through them.

**Step 3:** Identify key features:

- **Domain:** $(-\infty, +\infty)$
- **Range:** $(0, +\infty)$
- **y-intercept:** $(0, 1)$
- **Behavior:** Increasing function (since $3 > 1$)
- **Asymptote:** The x-axis ($y = 0$) is a horizontal asymptote as $x \to -\infty$
- **Special point:** Passes through $(1, 3)$

The graph rises steeply to the right and approaches zero on the left.

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### Example 5: Transformations of Exponential Functions

**Problem:** Describe how the graph of each function relates to the graph of $y = 2^x$:

a) $y = 2^{x+1}$
b) $y = 2^x + 3$
c) $y = -2^x$

**Solution:**

a)
$y = 2^{x+1}$
is a **horizontal shift** of
$y = 2^x$
by 1 unit to the left.

We can verify:
$2^{x+1} = 2 \cdot 2^x$
, so this is also equivalent to a vertical stretch by factor 2.

b)
$y = 2^x + 3$
is a **vertical shift** of
$y = 2^x$
by 3 units upward.

The horizontal asymptote moves from
$y = 0$
to
$y = 3$
.

c)
$y = -2^x$
is a **reflection** of
$y = 2^x$
across the x-axis.

The range changes from
$(0, +\infty)$
to
$(-\infty, 0)$
, and the function becomes decreasing.

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### Example 6: Solving Simple Exponential Equations

**Problem:** Solve for
$x$
:

a)
$2^x = 16$

b)
$3^{x-1} = 27$
c)
$(\frac{1}{2})^x = 8$

**Solution:**

a)
$2^x = 16$

Express 16 as a power of 2:
$2^x = 2^4$

Since the bases are equal:
$x = 4$

b)
$3^{x-1} = 27$

Express 27 as a power of 3:
$3^{x-1} = 3^3$

Since the bases are equal:
$x - 1 = 3$
, so
$x = 4$

c)
$(\frac{1}{2})^x = 8$

Express both sides as powers of 2:
$2^{-x} = 2^3$

Since the bases are equal:
$-x = 3$
, so

$x = -3$

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## Example 7: Exponential Growth Model

**Problem:** The population of a city was 500,000 in 2020 and is growing at a rate of 2.5% per year.

a) Write an exponential function
$P(t)$
that models the population
$t$
years after 2020.
b) Estimate the population in 2030.
c) In what year will the population reach 600,000?

**Solution:**

a) The exponential growth model is:
$$P(t) = P_0(1 + r)^t$$

where
$P_0 = 500{,}000$
is the initial population and
$r = 0.025$
is the growth rate.

$$P(t) = 500{,}000(1.025)^t$$

b) For the year 2030, we have
$t = 10$
:
$$P(10) = 500{,}000(1.025)^{10}$$

Using a calculator:
$(1.025)^{10} \approx 1.2801$

$$P(10) \approx 500{,}000 \times 1.2801 = 640{,}050$$

The population in 2030 will be approximately 640,000 people.

c) We need to solve:

$$500{,}000(1.025)^t = 600{,}000$$

Dividing both sides by 500,000:

$$(1.025)^t = 1.2$$

Taking logarithms (we'll learn this technique in detail later):

$$t = \frac{\ln(1.2)}{\ln(1.025)} \approx \frac{0.1823}{0.0247} \approx 7.38$$

The population will reach 600,000 approximately 7.38 years after 2020, which is during the year 2027.

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## Example 8: Exponential Decay Model

**Problem:** A radioactive substance has a half-life of 5 years. If we start with 100 grams, how much remains after 15 years?

**Solution:**

The exponential decay model for half-life is:

$$A(t) = A_0 \left(\frac{1}{2}\right)^{t/h}$$

where

$A_0$

is the initial amount,

$t$

is time, and

$h$

is the half-life.

Given:

$A_0 = 100$

grams,

$h = 5$

years,

$t = 15$

years

$$A(15) = 100\left(\frac{1}{2}\right)^{15/5} = 100\left(\frac{1}{2}\right)^{3} = 100 \cdot \frac{1}{8} = 12.5$$

After 15 years, 12.5 grams of the substance remains.

**Verification:** After 5 years: 50g. After 10 years: 25g. After 15 years: 12.5g. Each 5-year period halves the amount, which confirms our answer.

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## 🌍 Applications (Ứng dụng Thực tế)

### Application 1: Compound Interest

**Problem:** You invest $10,000 in a savings account with an annual interest rate of 6%, compounded monthly. How much money will you have after 5 years?

**Solution:**

The compound interest formula is:

$$A = P\left(1 + \frac{r}{n}\right)^{nt}$$

where:

- $P = 10{,}000$
  (principal)
- $r = 0.06$
  (annual interest rate)
- $n = 12$
  (compounding periods per year)
- $t = 5$
  (years)

$$A = 10{,}000\left(1 + \frac{0.06}{12}\right)^{12\times 5} = 10{,}000(1.005)^{60}$$

Using a calculator:

$$(1.005)^{60} \approx 1.3489$$

$$A \approx 10{,}000 \times 1.3489 = 13{,}489$$

After 5 years, you will have approximately $13,489.

**Comparison with simple interest:**

With simple interest:

$$A = P(1 + rt) = 10{,}000(1 + 0.06 \times 5) = 13{,}000$$

The compound interest yields an additional $489 due to "interest on interest."

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## Application 2: Population Growth

**Problem:** In 2020, Vietnam's population was approximately 97.34 million with an annual growth rate of 0.91%. Assuming this growth rate remains constant, estimate Vietnam's population in 2050.

**Solution:**

Using the exponential growth formula:

$$A = Pe^{rt}$$

where:

- $P = 97.34$
  million (initial population in 2020)
- $r = 0.0091$
  (growth rate as a decimal)
- $t = 30$
  years (from 2020 to 2050)

$$A = 97.34 \times e^{0.0091 \times 30} = 97.34 \times e^{0.273}$$

Using a calculator:

$$e^{0.273} \approx 1.3138$$

$$A \approx 97.34 \times 1.3138 \approx 127.88$$

Vietnam's population in 2050 is estimated to be approximately 127.88 million people.

**Note:** This assumes constant growth rate, which may not reflect reality due to changing birth rates, mortality rates, and migration patterns.

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## ✏️ Practice Exercises (Bài tập Luyện tập)

### Exercise 1: Identify which of the following are exponential functions and state the base:

a)
$y = 5^x$

b)
$y = x^5$

c)
$y = e^x$

d)
$y = \left(\frac{2}{3}\right)^x$

### Exercise 2: Evaluate without a calculator:

a)
$4^{3/2}$

b)
$9^{-1/2}$

c)
$\left(\frac{1}{8}\right)^{-2/3}$

### Exercise 3: Compare the following numbers without calculating their exact values:

a)
$5^{1.2}$
and
$5^{1.5}$

b)
$(0.7)^{0.3}$
and
$(0.7)^{0.5}$

**Exercise 4: Sketch the graph of**
$y = \left(\frac{1}{3}\right)^x$
**and identify its domain, range, and asymptote.**

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## 📋 Review Exercises (Bài tập Ôn tập)

### Exercise 1: Multiple representations

For the function
$f(x) = 2^x$
:

a) Complete the table:

| $x$ | $-2$ | $-1$ | $0$ | $1$ | $2$ | $3$ |
|---|---|---|---|---|---|---|
| $f(x)$ | ? | ? | ? | ? | ? | ? |

b) State the domain and range.
c) Is the function increasing or decreasing?
d) What is the y-intercept?

### Exercise 2: Exponential equations

Solve for
$x$
:

a)
$5^x = 125$
b)
$2^{2x-1} = 16$
c)
$\left(\frac{1}{4}\right)^x = 64$

### Exercise 3: Transformations

Describe how each graph relates to

$y = 3^x$

:

a)

$y = 3^{x-2}$

b)

$y = 3^x - 4$

c)

$y = 2 \cdot 3^x$

### Exercise 4: Real-world application

A car purchased for $25,000 depreciates at a rate of 15% per year.

a) Write an exponential function

$V(t)$

for the car's value after

$t$

years.

b) What will the car be worth after 4 years?

c) When will the car's value drop below $10,000?

### Exercise 5: Continuous compound interest

An investment of $5,000 earns interest at an annual rate of 4.5%, compounded continuously using the formula

$A = Pe^{rt}$

.

a) How much will the investment be worth after 10 years?

b) How long will it take for the investment to double?

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## 📝 Chapter Summary (Tóm tắt Chương)

### Key Concepts

1. **Exponential Function Definition**

- Form:
  $y = a^x$
  where
  $a > 0$
  ,
  $aeq1$

- The variable is in the exponent
- The base is constant

2. **Properties**
   - Domain:
     $\mathbb{R}$
     (all real numbers)
   - Range:
     $(0, +\infty)$
     (all positive numbers)
   - Continuous everywhere
   - No x-intercept, y-intercept at
     $(0, 1)$

   - Horizontal asymptote:
     $y = 0$

3. **Behavior Based on Base**
   - When
     $a > 1$
     : increasing function (exponential growth)
   - When
     $0 < a < 1$
     : decreasing function (exponential decay)
4. **The Natural Base e**
   - $e \approx 2.71828$

- $e = \lim_{n \to \infty} (1 + \frac{1}{n})^n$

- $y = e^x$
  is the natural exponential function

5. **Applications**
   - Compound interest:
     $A = P\left(1 + \frac{r}{n}\right)^{nt}$

   - Continuous growth/decay:
     $A = Pe^{rt}$

   - Population models
   - Radioactive decay
   - Depreciation

## Key Formulas

| Formula | Description |
| --- | --- |
| $y = a^x$ | Basic exponential function |
| $a^m \cdot a^n = a^{m+n}$ | Product rule for exponents |
| $\frac{a^m}{a^n} = a^{m-n}$ | Quotient rule for exponents |
| $(a^m)^n = a^{mn}$ | Power rule for exponents |
| $A = P(1 + r)^t$ | Compound interest (annual) |
| $A = Pe^{rt}$ | Continuous compound interest |
| $A = A_0\left(\frac{1}{2}\right)^{t/h}$ | Half-life decay model |

## Problem-Solving Strategies

1. **Identifying exponential functions:** Check if the variable is in the exponent
2. **Evaluating exponential expressions:** Use exponent rules and express in terms of known powers
3. **Comparing exponential values:** Use monotonicity properties based on the base
4. **Solving exponential equations:** Express both sides with the same base when possible
5. **Modeling real situations:** Identify initial value, growth/decay rate, and time period

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🎯 **Looking Ahead:** In the next chapter, we will explore **logarithmic functions**, which are the inverse functions of exponential functions. Logarithms will give us powerful tools for solving exponential equations and modeling phenomena where we need to find the exponent rather than the result.